

Specialisation:

Stochastic Modelling, Machine Learning and Wireless Networks.

EDUCATION

Télécom Paris (LTCI lab) - INRIA (MathNet team)

Mar 2025 - Mar 2028

Ph.D. in Applied Probability

Paris, France

- Subject: Stochastic Modelling of Non-Terrestrial Networks (NTNs).
- Areas: Stochastic geometry, ergodic theory, Palm calculus.
- Side activity: Part of ESA's NTN forum for TN-NTN integration and standards definition.
- Supervisors: Prof. F. Baccelli (ENS Paris), Prof. L. Decreusefond (Télécom Paris), Dr. N. Soprano-Loto (Inria).

École Polytechnique Fédérale de Lausanne (EPFL)

Apr 2024 - Sept 2024

Master's Thesis Visiting Student - Computer Science

Lausanne, Switzerland

- Thesis: Dynamic Bayesian Optimisation (DBO) for improving the performance of Cellular Networks.
- Key Results: Theorised a novel probabilistic mobility model and developed a cellular network simulator to evaluate DBO algorithm for downlink power optimisation against naïve approaches.
- Supervisors: Prof. P. Thiran (EPFL), Dr. A. Bardou (EPFL).

École Polytechnique

Sept 2022 - Sept 2024

MSc in Computer Science - Track: Internet of Things

Paris, France

- Double Major: Machine Learning, Computer Networks; GPA: 3.89/4.
- Key subjects: Machine, Reinforcement and Deep Learning, Federated Learning, Large scale optimization, Computer and Wireless Networks, Embedded Systems.

Linköping University (LIU)

Aug 2021 - Jan 2022

Exchange Programme - Industrial Engineering

Linköping, Sweden

- GPA: 4.5/5.

Alma Mater Studiorum – Università di Bologna (UNIBO)

Sept 2019 - Jul 2022

BSc in Industrial Engineering - Minor: Information & Communication Technology

Bologna, Italy

- Final mark: 110/110 cum LAUDE; GPA: 28.94/30.
- Thesis: Conception and Design of a Predictive Maintenance System for the Automotive Sector.
- Key subjects: Signal Processing, Operation Research, Electronics, Logistics, Stochastic Process Control.

WORK EXPERIENCE

INRIA (MATHNET team)

Sept 2024 - Mar 2025

R&D Engineer - Predoctoral

Paris, France

- Research as part of the EU-funded INSTINCT project, applying stochastic geometry and filtering theory for modelling Joint Communication and Sensing (JCAS) scenarios in 6G networks.

- Pursued academic training on advanced probability theory and satellite communications.
- Supervisor: Prof. F. Baccelli (ENS Paris).

Cisco Meraki

Apr 2023 - Sept 2023

Software Engineer Intern - Data Science team

Paris, France

- Theorised and developed a novel time series prediction model and a causal inference algorithm. Both algorithms were used to build an anomaly detection product for L2 devices.
- Key Results: Conference publication [1].
- Supervisors: Prof. T. H. Clausen (École Polytechnique) and Dr. Z. Yao (Cisco Meraki).

National Inter-University Consortium for Telecommunications (CNIT)

Nov 2022 - Apr 2023

Scientific Research Partner

Bologna, Italy

- Developed a low computational cost algorithm for RIS-aided UAV tracking in a non-line-of-sight scenario.
- Key Results: Contribution to publication [2].
- Supervisor: Prof. F. Guidi (UNIBO).

Cluster Reply

Mar 2022 - May 2022

Junior IT Consultant Intern

Turin, Italy

- Developed a predictive maintenance algorithm for vehicles failure prediction, combining statistical process control and regression-based techniques.
- Jointly led sales and product development.
- Key Results: Algorithm got acquired by third-part company (Movyon) for Tunnel Fault Monitoring.
- Supervisors: Prof. F. Guidi (UNIBO) and S. Agostini, Associate Partner (Cluster Reply).

Junior Enterprise Bologna (JEBO)

Jul 2021 - Jul 2022

Vice President

Bologna, Italy

- Managed 90 human resources, overseeing 51K € in revenue (+60%).
- Led partnership management – created 10 new collaborations.
- Creator of “Coffee with Entrepreneurs” career orientation program (moderator of 3 events):
 - 30 entrepreneurs involved, 1K podcast views, avg. 30 attendees per event.

PUBLICATIONS

- [1] E. Mengoli, Z. Yao, W. Wei, and T. Clausen, “Develop end-to-end anomaly detection system,” in *2023 IEEE International Conference on Data Mining Workshops (ICDMW)*, IEEE, 2023, pp. 1370–1379.
- [2] F. Guidi, A. Guerra, E. Mengoli, and A. Zanella, “A statistical characterization of communication performance in ris-aided networks,” in *ICASSP 2024-2024 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, IEEE, 2024, pp. 12 921–12 925.
- [3] E. Mengoli, L. Moll, V. Strozzi, and E.-M. El-Mhamdi, “On the byzantine fault tolerance of signsgd with majority vote,” *arXiv preprint arXiv:2502.19170*, 2025.

INVITED TALKS

13ème Atelier en Évaluation des Performances

Dec, 2024
Toulouse, France

MathNet Reading group

Apr, 2026
Paris, France

CONTRIBUTED TALKS

EuCNC & 6G Summit
Demo booth presentation.

June, 2025
Poznań, Poland

5th Italian Meeting on Probability and Mathematical Statistics
Speaker and organizer of a contributed session.

June, 2026
Palermo, Italy

Stochastic Networks Conference
Poster session.

June, 2026
Chicago, USA

RESEARCH SCHOOLS

Random walks: applications and interactions

Jan, 2026
CIRM, Marseille, France

XXIX Brazilian School of Probability
Poster session.

August, 2026
Rio de Janeiro, Brazil

EXTRACURRICULAR ACTIVITY

Federazione Italiana Pallacanestro (FIP)
Basketball referee

Dec 2014 - Sept 2022
Italy

- “Trofeo delle regioni” (2019) - selected as one of the two best referees in Tuscany.
- “14th World Maxibasketball Fimba Championship” (2017).

SKILLS SUMMARY

- **Languages:** Italian: Native, English: Proficient, Spanish: Fluent, French: Fluent.
- **Tools:** Python, SQL, Java, C, LaTeX, AWS, Azure, Docker, Git, GitLab, GitHub, MySQL, Arduino, Trello, Jira, Microsoft Office Suite, Google Suite, Google Data Studio.